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LANGTON'S ANT



Langton's Ant is a cellular automata simulation on a two-dimensional grid, similar to Project 13, "Conway's Game of Life." In the simulation, an "ant" begins on a square that is one of two colors. If the space is the first color, the ant switches it to the second color, turns 90 degrees to the right, and moves forward one space. If the space is the second color, the ant switches it to the first color, turns 90 degrees to the left, and moves forward one space.

Despite the very simple set of rules, the simulation displays complex emergent behavior. Simulations can feature multiple ants in the same space, causing interesting interactions when they cross paths with each other. Langton's Ant was invented by computer scientist Chris Langton in 1986. More information about Langton's Ant can be found at https://en.wikipedia.org/wiki/Langton%27s_ant.